

1. Knowledge Framework / 知识框架

Part I: Low-Level Vision / 低层视觉 (lec00–lec05)

Lecture	Topic / 主题
lec00	Introduction to Computer Vision / 计算机视觉导论
lec01	Images & Image Filtering / 图像与图像滤波
lec02	Edge Detection / 边缘检测
lec03	Image Resampling & Interpolation / 图像重采样与插值
lec04	Harris Corner Detection / Harris角点检测
lec05	Feature Invariance / 特征不变性

Part II: Geometry & Appearance / 几何与外观 (lec06–lec10, lec13–lec14)

Lecture	Topic / 主题
lec06	Feature Descriptors & Matching / 特征描述子与匹配
lec07	Image Transformations & Warping / 图像变换与变形
lec08	Image Alignment / 图像对齐
lec09	RANSAC / 随机抽样一致
lec10	Cameras & Projection / 相机与投影
lec13	Binocular Stereo / 双目立体视觉
lec14	Multi-View Stereo / 多视图立体

Part III: Recognition & Deep Learning / 识别与深度学习 (lec15–lec16)

Lecture	Topic / 主题
lec15	Neural Networks / 神经网络
lec16	CNNs, RNNs, Attention & Transformers / 卷积网络, 循环网络, 注意力与 Transformer

Part IV: Generative Models / 生成模型 (下半学期)

Lecture	Topic / 主题
lecure4	Maximum Likelihood & MLE / 最大似然估计
lecure5	Latent Variable Models & ELBO / 潜变量模型与证据下界
LVM_02	VAE & Reparameterization / 变分自编码器与重参数化
NF_01	Normalizing Flow Models (1) / 标准流模型(1)
lecure8	Normalizing Flow Models (2) / 标准流模型(2)
lecure9	GANs / 生成对抗网络
lec10	F-GANs, Wasserstein GAN, BIGAN, CycleGAN, F-数度GAN, WGAN, 双向GAN, 循环GAN
lec11	Energy-Based Models / 基于能量的模型
lec12	Score Matching & NCE / 分数匹配与噪声对比估计
lec13	Score-Based Models (1) / 基于分数的模型(1)
lec14	Score-Based Models (2) / 基于分数的模型(2)
lec15	Diffusion Models (DDPM, SDE) / 扩散模型
lec16	Diffusion Models: SDE+ODE, Architecture / 扩散模型进阶

2. Core Concepts / 核心概念梳理

2.1 Image Filtering / 图像滤波

Convolution / 卷积

- Definition (定义): Replace each pixel with a weighted sum of its neighbors using a kernel. / 用卷积核将每个像素替换为加权的加积和。
- Formula (公式): $G[i,j] = \sum_{m,n} F[m,n] \cdot \text{kernel}[j-m, i-n]$
- Key property (关键性质): Convolution is associative and commutative; convolution with itself is another Gaussian. / 卷积满足结合律和交换律。

Gaussian Filter / 高斯滤波器

- Definition (定义): A low-pass filter that removes high-frequency components. / 去除高频成分的平滑滤波器。
- Property (性质): Convoluting two Gaussians of σ_1 and σ_2 yields a Gaussian of $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$. / 两个高斯卷积仍为高斯。

Sharpening Filter / 锐化滤波器

- Definition (定义): original + (original - blurred) = $\nabla^2 \cdot \text{convolution}(\text{original}) - \text{blurred}$. / 原图减去(原图减模糊图)实现锐化。
- Also called (亦称): High-pass filter / 高通滤波器。

2.2 Edge Detection / 边缘检测

Image Gradient / 图像梯度

- Definition (定义): $\nabla f(x,y) = [\partial f / \partial x, \partial f / \partial y]^T$, points in direction of most rapid increase. / 指向强度增长最快的方向。
- Gradient magnitude (梯度模值): $|\nabla f| = \sqrt{(\partial f / \partial x)^2 + (\partial f / \partial y)^2}$
- Gradient direction (梯度方向): $\theta = \arctan(\partial f / \partial y / \partial f / \partial x)$

Sobel Operator / Sobel算子

- Definition (定义): Common approximation of derivative of Gaussian. / 高斯导数的常用近似。
- Kernels (核):
 - x-direction: $[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]$
 - y-direction: $[1, 2, 1], [0, 0, 0], [1, -2, -1]$

Canny Edge Detector / Canny边缘检测器

1. Filter image with derivative of Gaussian / 用高斯导数滤波
2. Find magnitude and orientation of gradient / 求梯度模值和方向
3. Non-maximum suppression / 非极大值抑制
 - Thresholding (two thresholds): high T, low t / 滞后阈值化
5. Strong edges: R > T; Weak edges: $t < R < T$ (keep only if connected to strong edge) / 强边缘保留, 弱边缘仅在连与强边缘时保留

Scale Space / 尺度空间

- Properties (性质): Edge position may shift with σ ; edges may merge; an edge cannot split into two. / 边缘位置可能随 σ 偏移, 可合并但不能分裂。

2.3 Image Resampling / 图像重采样

Aliasing / 混叠

- Definition (定义): Occurs when sampling rate < 2 * max frequency (Nyquist rate). / 采样率低于奈奎斯特率时发生。
- Solution (解决方案): Gaussian pre-filtering before subsampling. / 下采样前先做高斯预滤波。

Gaussian Pyramid / 高斯金字塔

- Definition (定义): Repeatedly blur and downsample by 2x. / 反复模糊后2倍下采样。
- Space (空间): Total space = $\sqrt{4/3}$ * original image. / 总空间为原图的 $\sqrt{4/3}$ 。

Interpolation Methods / 插值方法

- Nearest-neighbor (最近邻): Fastest, but blocky. / 最快但产生方块效应。
- Bilinear (双线性): Better quality, tent function. / 质量更好。
- Bicubic (双三次): Best quality among common choices. / 常用方法中质量最好。

2.4 Harris Corner Detection / Harris角点检测

SSD Error / SSD误差

- Formula (公式): $E(u,v) = \sum_{i,j} w(i,j) \cdot w(x,y) \cdot |I(x+u, y+v) - I(x,y)|^2$
- Small motion approximation (小运动近似): $E(u,v) \approx \sum_{i,j} w(i,j) \cdot [u \cdot I_x + v \cdot I_y]$, where H is the second moment matrix. / 二阶矩矩阵近似。

Second Moment Matrix / 二阶矩矩阵

- Definition (定义): $H = \sum w(x,y) \cdot \begin{bmatrix} x^2 & x \cdot y \\ x \cdot y & y^2 \end{bmatrix}$
- Eigenvalues (特征值): λ_1, λ_2 determine corner/edge/flat classification. / 决定角点/边缘/平坦区域分类。

Classification / 分类

Case (情况)	λ_1, λ_2	Classification (分类)
Both small (都很小)	$\lambda_1 \approx \lambda_2 \approx 0$	Flat / 平坦区域
One large, one small (一大一小)	$\lambda_1 \gg \lambda_2 \approx 0$	Edge / 边缘
Both large (都很大)	$\lambda_1 \gg \lambda_2 \gg 0$	Corner / 角点

Harris Operator / Harris算子

- Formula (公式): $S = \det(H) - \frac{1}{4} (\text{trace}(H))^2 = \lambda_1 \lambda_2$
- Typical Steps (典型步骤):
 - $\Delta x = 0.06 \sigma$

Harris Detector / Harris检测步骤

1. Compute image gradients / 计算图像梯度
2. Square of derivatives, Gaussian filter / 梯度平方后高斯滤波
3. Compute cornerness function / 计算角点响应函数
4. Non-maxima suppression / 非极大值抑制

2.5 Feature Invariance / 特征不变性

- Invariance (不变性): Features locations don't change after transformation. / 变换后特征位置不变。
- Equivariance (等变性): Features detected at corresponding locations in transformed images. / 变换后特征在对应位置被检测到。

Harris Invariance Properties / Harris不变性

Transformation (变换)	Property (性质)
Translation (平移)	Invariant / 不变
Rotation (旋转)	Equivariant (eigenvalues unchanged) / 等变
Affine Invariant (仿射不变)	Partially invariant / 部分不变
Scale (尺度)	NOT invariant / 不变

Scale Invariant Detection / 尺度不变检测

- Key idea (核心思想): Find local maximum in both position AND scale. / 在位置 and 尺度上同时寻找局部极大值。
- Laplacian of Gaussian (LoG): Blob detector, find maxima in position-scale space. / 斑点检测器, 在位置-尺度空间找极大值。
- Difference of Gaussians (DoG): Approximation of LoG, used in SIFT. / LoG的近似, SIFT中使用。

Characteristic Scale / 特征尺度

- Definition (定义): Scale producing peak LoG response. / 产生LoG峰值响应的尺度。

2.6 Feature Descriptors & Matching / 特征描述子与匹配

SIFT Descriptor / SIFT描述子

- Process (过程):
 - Take $3 \times 3 \times 3$ window around feature / 取 $3 \times 3 \times 3$ 窗口
 - Compute gradient orientation per pixel / 计算每像素梯度方向
 - Threshold weak edges / 阈值化弱边缘
 - Create $4 \times 4 \times 8$ grid of cells, 8 orientation bins each / $4 \times 4 \times 8$ 网格, 每格8方向bin
 - Result: 128-dimension descriptor. / 结果为128维描述子
- Properties (性质): Robust to viewpoint (up to $\sim 60^\circ$), illumination changes. / 对视点变化 ($\sim 60^\circ$ 以内) 和光照变化鲁棒。

HOG Descriptor / HOG描述子

- Steps: Gradient computation $\rightarrow$ rightarrow Cell histograms (9 bins) $\rightarrow$ rightarrow Block normalization (L2-norm). / 梯度计算 $\rightarrow$ 块直方图 $\rightarrow$ 块归一化。
- ORB Descriptor / ORB描述子
 - Steps: oFAST keypoint detection $\rightarrow$ rightarrow rBRIEF binary descriptor (256 bits). / oFAST关键点检测 $\rightarrow$ rBRIEF二进描述子。

Feature Matching / 特征匹配

- L2 distance: $\|f_1 - f_2\|_2$
- Ratio distance: $\|f_1 - f_2\|_1 / \|f_1\|_1 + \|f_2\|_1$ ($\|f_2\|_1 = \text{best match}, \|f_2\|_1 = 2 \cdot \text{best}$) / 比值得距离
- ROC Curve: True positive rate vs. false positive rate; AUC summarizes performance. / ROC曲线与AUC指标。

2.7 Image Transformations / 图像变换

Transformation Invariant / 变换不变性

Transform (变换)	Matrix (矩阵)	DOF	Properties (性质)
Translation (平移)	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	2	Origin may shift / 原点可移动
Euclidean (欧氏)	$R + t$	3	Preserves distances / 保持距离
Similarity (相似)	$sR + t$	4	Preserves angles / 保持角度
Affine (仿射)	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	6	Parallel lines preserved / 平行线保持
Projective (射影/单应)	General 3x3	8	Lines preserved, parallel not / 直线保持, 平行不一定

Homography / 单应变换

- Definition (定义): 8-DOF projective transformation; defined up to scale. / 8自由度射影变换, 由尺度因子决定。
- Minimum matches (最少匹配): 4 point correspondences. / 4对匹配点。

Forward vs. Inverse Warping / 前向与逆向变形

- Forward warping: Send source pixels to target; may cause holes. / 源像素送到目标, 可产生空洞。
- Inverse warping: For each target pixel, sample from source via T^{-1} ; use interpolation. / 对每个目标像素从源采样, 需插值。

2.8 Image Alignment & Least Squares / 图像对齐与最小二乘

Least Squares / 最小二乘法

- Translation (平移): $t^* = \text{mean displacement}$; normal equations: $A^T A \cdot t = A^T b$.
- Affine (仿射): 6 unknowns, need ≥ 3 matches. / 6个未知量, 至少3对匹配。
- Homography (射影): Solve via SVD; h = eigenvector of $A^T A$ with smallest eigenvalue. / 通过SVD求解。

2.9 RANSAC / 随机抽样一致

Algorithm / 算法步骤

1. Randomly select s samples (minimum to fit model) / 随机选 s 个样本
 2. Fit model / 拟合模型
 3. Count inliers (within threshold Δ) / 计数内点
 4. Repeat N times / 重复N次
 5. Choose model with most inliers / 选择内点最多的模型
 6. Refit using all inliers (least squares) / 用所有内点重新拟合
- Number of Rounds / 迭代次数
- Formula (公式): $SN = \frac{1}{1 - \text{Vrac}(\log(1-p)) \log(1-(1-p)^N)}$
 - p = desired probability of success / 期望成功概率
 - e = outlier ratio / 外点比例
 - s = minimum sample size of smallest samples

Pros & Cons / 优缺点

- Cons: Simple, general, often works well. / 简单、通用、效果好。
- Cons: Parameters to tune; can fail for very low inlier ratios. / 需调参: 内点率极低时可能失败。

2.10 Cameras & Projection / 相机与投影

Pinhole Camera / 针孔相机

- Projection (投影): $(x,y,z) \rightarrow (f \cdot \frac{x}{z}, f \cdot \frac{y}{z})$ using similar triangles. / 利用相似三角形投影。
- Perspective projection matrix / 透视投影矩阵: $K = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$

Camera Parameters / 相机参数

- Intrinsic (内参): focal length f , principal point (c_x, c_y) , aspect ratio α , skew s .
- Extrinsic (外参): Rotation R and translation t .
- Full projection (完整投影): $x_c = K \cdot [R | t] \cdot X_w$
- Perspective (透视): General case, parallel lines converge. / 一般情况, 平行线汇聚。
- Orthographic (正交): Distance to image plane = ∞ .
- Weak perspective (弱透视): Scaled orthographic; good for small depth range. / 缩放正交投影。

Radial Distortion / 径向畸变

- Types: Barrel (桶形) vs. Pincushion (枕形)
- Model: $r_{\text{corrected}} = r(1 + k_1 r^2 + k_2 r^4 + \dots)$

2.11 Stereo Vision / 立体视觉

Disparity & Depth / 视差与深度

- Disparity (视差): $z_d = x_2 - x_1$ (horizontal pixel difference between views). / 左右视图水平像素差。
- Depth (深度): $z = f \cdot B / d$, where B = baseline, f = focal length. / 深度与视差成反比。

Stereo Matching / 立体匹配

- SSD matching: SSD = squared differences between a window. / 窗口内平方差之和。
- Window size tradeoff: Small $\rightarrow$ more detail but more noise; Large $\rightarrow$ less noise but loss of detail. / 窗口大小的权衡。

Stereo as Energy Minimization / 立体匹配作为能量最小化

- Energy: $E(d) = \sum \text{smoothness_cost}(d(x), d(y)) + \lambda \sum \text{data_cost}(d(x), d(y))$
- Smoothness costs: Potts model (if same, λ diff; if different, $\lambda \cdot 1$ distance. / Potts模型。

Multi-View Stereo (MVS) / 多视图立体

- SSDS: Sum of SSD over multiple baselines. / 多基线SSD之和。
- Matching scores: SSD, SAD, NZCC (zero-mean normalized cross-correlation). / 匹配评分方法。

Plane-Sweep Stereo / 平面扫描立体

- Sweep family of planes at different depths, reproject neighbors, compare. / 在不同深度扫描平面, 重投影邻居并比较。
- NeRF: Represent scenes as $S(x,y,z) \rightarrow \text{volume rendering}$. / 用神经网络表示场景的辐射场。

2.12 Neural Networks / 神经网络

Linear Classifier / 线性分类器

- Score function (得分函数): $f(x) \cdot W = Wx + b$
- Three viewpoints (三个视角): Algebraic (代数), Geometric (几何), Template matching (模板匹配)

Loss Functions / 损失函数

- Cross-entropy loss (叉熵损失): $J = -\log(\exp(f_{ij}) / \sum \exp(f_{ij}))$
- Softmax (最大熵): $P(y=j) = \exp(f_j) / \sum \exp(f_j)$

Neural Network Architecture / 神经网络架构

- MLP (多层感知机): Linear functions separated by non-linear activations. / 线性函数间插非线性的激活函数。
- Gradient descent (梯度下降): $\nabla W \leftarrow -\eta \nabla L(W)$; learning rate η crucial. / 学习率 η 至关重要。
- Backpropagation (反向传播): Efficient gradient computation via chain rule. / 通过链式法则高效计算梯度。

2.13 CNNs, RNNs, Attention, Transformers / CNN. RNN. 注意力. Transformer

Convolutional Layer / 卷积层

- Output size (输出尺寸): $(N - F) \cdot \text{stride} + 1$
- Filters (滤波器): Extend full depth of input; produce activation maps. / 滤波器覆盖输入全深度, 产生激活图。
- Padding & Stride: Control output spatial dimensions. / 控制输出空间维度。

Pooling Layer / 池化层

- Max pooling (最大池化): Reduces spatial dimensions; operates per activation map. / 减小空间维度, 逐图池化操作。

RNN / 循环神经网络

- Recurrence (递推): $h_t = f(W \cdot \text{input}_t + U \cdot h_{t-1})$; same weights at each time step. / 每个时间步使用相同权重。
- BPTT: Backpropagation Through Time. / 时间反向传播。
- Truncated BPTT: Backpropagate only for limited steps. / 截断时间反向传播。
- Types: Vanilla RNN, LSTM, GRU. / 类型。

Self-Attention / 自注意力

- Query-Key-Value (查询-键-值): $\text{Attention}(Q,K,V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$
- Multi-Head Attention (多头注意力): Multiple attention heads in parallel; concatenate outputs. / 多个注意力头并行, 拼接输出。

Transformer

- Encoder (编码器): Self-attention + FFN, with residual connections and layer norm. / 自注意力+前馈网络, 残差连接并层归一化。
- Decoder (解码器): Masked multi-head attention (cannot see future) + cross-attention + FFN. / 掩码多头注意力+交叉注意力+前馈网络。
- Positional Encoding (位置编码): Injects sequence order information. / 注入序列顺序信息。

2.14 Maximum Likelihood Estimation / 最大似然估计

KL Divergence / KL散度

- Definition (定义): $D_{KL}(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx$
- Asymmetric (不对称): 衡量两个分布的差别。
- Interpretation (解释): Extra bits needed when using code optimized for q on data from p . / 用 q 的编码压缩来自 p 的数据时多用的比数。

MLE / 最大似然估计

- Formula (公式): $\hat{\theta} = \arg\max_{\theta} \sum \log p(x|\theta)$
- Equivalence (等价性): Minimizing $\sum \log p(x|\theta)$ is equivalent to maximizing $\sum \log \frac{1}{p(x|\theta)}$ = maximizing expected log-likelihood. / 最小化KL散度等价于最大化期望对数似然。

Monte Carlo Estimation / 蒙特卡罗估计

- Unbiased (无偏): $E[\hat{\theta}] = \theta$
- Variance (方差): $\text{Var}(\hat{\theta}) \propto 1/N$; reduces with more samples. / 更多样本降低方差。

Bias-Variance Tradeoff / 偏差-方差权衡

- High bias (高偏差): Underfitting (model too simple). / 欠拟合。
- High variance (高方差): Overfitting (model too complex). / 过拟合。
- Regularization (正则化): $\text{loss} = \text{loss} + \lambda \|w\|^2$. / 目标函数加入正则项。

2.15 Latent Variable Models & VAE / 潜变量模型与变分自编码器

Latent Variable Models / 潜变量模型

- Generative process (生成过程): $z \sim p(z), x \sim p(x|z)$. / 先采样潜变量, 再由其生成数据。
- Marginal likelihood (边际似然): $p(x) = \int p(x|z)p(z) dz$ — often intractable. / 通常难以计算。

Mixture of Gaussians / 高斯混合模型

- $z \sim \text{mixture}(\text{Categorical}(\{1, \dots, K\}), \text{Gaussian}(\mu_k, \sigma_k))$ / 离散潜变量的混合模型。

VAE / 变分自编码器

- Infinite mixture (无限混合): $z \sim \text{mixture}(\text{Normal}(0, I), p(z))$ where $p(z)$ are neural networks. / 连续潜变量的无限混合高斯。

ELBO / 证据下界

- Formula (公式): $\log p(x|\theta) \geq \mathbb{E}_{q(z|\theta)} [\log p(x|z)] + H(q)$
- Equality condition (等号条件): When $q(z) = p(z|\theta)$, i.e., $\text{KL}(q||p) = 0$.
- Decomposition (分解): $\log p(x) = \text{ELBO} + D_{KL}(q||p)$

Autoencoder Perspective / 自编码器视角

- ELBO = Reconstruction term - KL term: $\text{ELBO} = \mathbb{E}[\log p(x|z)] - \text{KL}(q||p)$
- Reconstruction (重建项): Encourages $\hat{x} \approx x$. / 鼓励重建接近原始输入。
- KL term (KL项): Forces latent distribution close to prior $p(z)$. / 迫使潜变量分布接近先验。

Reparameterization Trick / 重参数化技巧

- Key idea (核心思想): Sample $\epsilon \sim \text{Normal}(0, I)$, then $z = \mu + \sigma \epsilon$. / 将随机机转移到 ϵ , 使梯度可通过确定性函数传递。
- Benefits (优势): Enables gradient-based optimization; lower variance than REINFORCE. / 实现梯度优化, 方差低于REINFORCE。

Amortized Inference / 摊销推理

- Idea (思想): Learn $q_\phi(z|x)$ to map each x to variational parameters. / 学习编码器网络将 x 映射到变分参数, 避免逐点优化。

2.16 Normalizing Flow Models / 标准流模型

Core Idea / 核心思想

- Make $q = p$ deterministic and invertible; then $p(x)$ is tractable via change of variables. / 使映射确定且可逆, 通过变量替换公式计算似然。

Change of Variables / 变量替换公式

- ID: $p_X(x) = p_Z(z(x)) |J|$, where $J = \frac{dz}{dx}$. / 一维情况。
- General: $p_X(x) = p_Z(z(x)) |\det(\frac{\partial z}{\partial x})|$

Flow of Transformations / 变换流

- Composition (组合): $z_0 \rightarrow z_1 \rightarrow z_2$
- Likelihood: $p_X(x) = p_Z(z_0) |\det \frac{\partial z_0}{\partial x}|$

- Gradient penalty (梯度惩罚, WGAN-GP): Add $\lambda \text{norm}(\nabla \text{hat}(x))\{\|\nabla \text{hat}(x)\|_2 - 1\}^2$ to the objective. More stable / 更稳定的梯度惩罚方法。
- Advantages over original GAN (相比原始GAN的优势):
 - Loss correlates with sample quality / 损失与样本质量相关
 - More stable training / 训练更稳定
 - No mode collapse in practice / 实践中无模式坍塌

2.20 GAN / 双向生成对抗网络

Bidirectional GAN (BiGAN) / 双向GAN

- Motivation (动机): Standard GANs learn a generator $G: z \rightarrow x$ but cannot infer the latent representation z given x s. / 标准GAN只学到生成器，无法推断潜在表示。
- Architecture (架构): Add an encoder $E: x \rightarrow z$ alongside the generator $G: z \rightarrow x$. discriminator D takes $\{x, z\}$ pairs and distinguishes $G, E(x)$ from $G(z), z$ s. / 增加编码器 E ，判别器在 $\{x, z\}$ 对上区分编码对与生成对。
- Objective (目标): $\min_G [\mathbb{E}_x \text{vmax}_z \text{D}(V, E, G)] = \text{vminb}(\mathbb{E}_x [\text{vlim}_p \text{mathrm{d}atall}]) \log D(x, E(x)) + \text{vlimb}(\mathbb{E}_z [\text{vlim}_p \text{p}(\text{z})] \log(1 - D(G(z), z)))$.
- Result (结果): At equilibrium, $\mathbb{E}E = G^{-1}$ (encoder inverts the generator). / 均衡时编码器为生成器的逆。
- (Adversarially Learned Inference): Simultaneous proposal, equivalent to BiGAN. / 同时提出的等价方法。

2.21 CycleGAN / 循环生成对抗网络

Image-to-Image Translation / 图像到图像翻译

- Problem (问题): Given images from two domains X and Y , learn to translate between them / 给定两个域的图片，学习它们之间的翻译。
- Paired vs. Unpaired (配对与非配对): Paired examples are expensive; CycleGAN works with unpaired data. / 配对数据昂贵，CycleGAN使用非配对数据。

CycleGAN Architecture / CycleGAN架构

- Generators (生成器): $G_X: X \rightarrow Y$ and $G_Y: Y \rightarrow X$. / 双向生成器。
- Discriminators (判别器): D_X vs D_Y distinguishes real Y s from $G_X(X)$, D_Y distinguishes real X s from $G_Y(Y)$. / 双向判别器。
- Adversarial loss (对抗损失): $\mathbb{E}[\text{vlimb}(\mathbb{E}_x [\text{vlimb} \text{mathrm{GAN}}])G(D, Y, X, Y)] + \text{vlimb}(\mathbb{E}_x [\text{vlimb} \text{mathrm{GAN}}](F, D, X, X, Y))$. / 双向对抗损失。
- Cycle consistency loss (循环一致性损失): $\lambda \text{vlimb}(\text{mathbb{E}}_x [X(F(G(X)) - X)] + \text{vlimb}(\mathbb{E}_y [Y(G(Y)) - Y])]$. / 确保双向翻译一致。
- Overall objective (总体目标): $\min_{G, D} [\text{vlim}(\text{F}, G, D, X, D_Y)] + \text{vlimb}(\mathbb{E}_x [\text{vlimb} \text{mathrm{GAN}}]) + \lambda \text{vlimb} \text{vlimb}(\mathbb{E}_x [\text{vlimb} \text{mathrm{cycle}}])$.

AlignFlow / 齐流

- Key idea (核心思想): If $\mathbb{S}$ is a normalizing flow model, $\mathbb{S}^{-1}$ is free; exactly cycle-consistent. / 若生成器为流模型，正向自然得证，精确循环一致。
- Training (训练): Can train via MLE and/or adversarial learning. / 可通过MLE和/或对抗学习训练。

StarGAN / 星形GAN

- Extension (扩展): Handles multiple domains with a single generator. / 单一生成器处理多个域的翻译。

2.22 Energy-Based Models / 基于能量的模型

EBM Formulation / EBM公式

- Definition (定义): $\mathbb{S}_\theta(x) = \text{vfrac}(\text{vexp}(\text{f}_\theta(x))\{\text{Z}(\theta)\}) = \text{vfrac}(\text{1})\{\text{Z}(\theta)\}$ $\text{vexp}(\text{f}_\theta(x))$, where $\text{f}_\theta(x)$ is any function (e.g., neural network) and $\text{Z}(\theta) = \int \text{vlim} \text{vexp}(\text{f}_\theta(x))dx$ is the partition function. / 任意函数取指数后归一化，分母为配分函数。
- Key property (关键性质): Very flexible architecture — f_θ can be any neural network (no invertibility, factorization, or latent variable constraints). / 架构极其灵活， f_θ 可任意神经网络。
- Partition function (配分函数): $\text{Z}(\theta) = \int \text{vlim} \text{vexp}(\text{f}_\theta(x))dx$ — generally intractable. / 通常不可计算。

Energy Function / 能量函数

- Interpretation (解释): $\text{f}_\theta(x)$ is the “energy”; low energy = high probability. / 负能量函数为“能量”，低能量=高概率。
- Non-negativity trick (非负技巧): Given any function f_θ , ensure non-negativity via $\text{f}_\theta(x) = \text{f}_\theta(x) \wedge 2^5$, $\text{vexp}(\text{f}_\theta(x))$, S_θ . / 通过平方、指数，绝对值来保证非负。

Maximum Likelihood Training / 最大似然训练

- Log-likelihood (对数似然): $\log p_\theta(x) = \text{f}_\theta(x) - \log \text{Z}(\theta)$. / 对数似然等于函数值减去对数配分函数。
- Gradient (梯度): $\nabla \text{vlimb}(\mathbb{E}_x [\text{vlimb} \text{mathrm{train}}]) = \nabla \text{vlimb}(\text{f}_\theta(x) - \text{f}_\theta(x)) = \nabla \text{vlimb}(\text{f}_\theta(x))$. / 梯度=正相值 (推高训练数据能量) 减负值 (拉低模型采样能量)。
- Positive phase vs. Negative phase (正相位与负相位):
 - Positive phase (正相位): $\nabla \text{vlimb}(\text{f}_\theta(x))$ increases probability of training data. / 提高训练数据概率。
 - Negative phase (负相位): $\nabla \text{vlimb}(\text{f}_\theta(x))$ decreases probability of model samples. / 降低模型样本概率。

Contrastive Divergence / 对比数度

- Algorithm (算法): Instead of computing $\text{Z}(\theta)$ exactly, use Monte Carlo estimate. / 用蒙特卡洛估计代替精确计算配分函数。
 - Update step (更新步骤): $\text{f}_\theta(x) \leftarrow \text{f}_\theta(x) + \text{vlimb}(\text{f}_\theta(x)) - \text{vlimb}(\text{f}_\theta(x))$. / 用模型采样的梯度近似负值。
 - Key idea (核心思想): Make training data more likely than typical samples from the model. / 使训练数据相比模型典型样本概率更高。
- Sampling from EBMs / EBM采样**
- MCMC (马尔可夫链蒙特卡罗):
 - Initialize $\mathbb{S}_\theta$ randomly / 随机初始化
 - Propose $\mathbb{S}_\theta \leftarrow \mathbb{S}_\theta + \text{vlimb}(\text{noise})$ / 提议新样本
 - If $\text{f}_\theta(\mathbb{S}_\theta) > \text{f}_\theta(\mathbb{S}_\theta)$: accept; else accept with probability $\text{vexp}(\text{f}_\theta(\mathbb{S}_\theta) - \text{f}_\theta(\mathbb{S}_\theta))$ / 接受或一定概率接受
 - Converges to S_θ . / 收敛到 S_θ 。
 - Langevin MCMC (朗之万MCMC): $\mathbb{S}_\theta \leftarrow \mathbb{S}_\theta + \text{vlimb}(\text{noise}) + \text{vlimb}(\text{f}_\theta(\mathbb{S}_\theta))$. / 利用梯度信息加速采样的MCMC方法。
- Score Function (分数函数) for EBMs:** $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 配分函数为 $\text{Z}(\theta)$ 。
- Score function (分数函数) for EBMs: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 配分函数为 $\text{Z}(\theta)$ 。
 - Since $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$, / EBM的配分函数为 $\text{Z}(\theta)$ 。

Score-Based Generative Modeling / 基于分数的生成模型

- Key idea (核心思想): Directly estimate the score function $\text{S}_\theta(x)$ using a neural network $\text{vlimb}(\text{f}_\theta(x))$. / 直接用神经网络估计分数函数。
- Score matching (分数匹配): Train S_θ by minimizing $\mathbb{E}[\text{vlimb}(\text{mathrm{train}})]$. / 通过最小化训练损失来训练。
- Denoising Score Matching (去噪分数匹配): Add noise S_θ to data, then match the score of the perturbed distribution. / 对数据加噪声后匹配扰动分布的分数。
- Noise Conditional Score Network (NCN): Score function $\text{S}_\theta(x, z)$. / 使用多尺度噪声，训练噪声条件分数网络。
- Annealed Langevin Dynamics (退火动力学): Sequentially sample with decreasing noise levels; each level initialized from previous. / 逐步降低噪声水平采样。

EBM vs. Other Generative Models / EBM与其他生成模型比较

- Vs. Autoregressive/Flow/VAE: EBM has much more flexible architecture (no restrictions on f_θ), but sampling is slower (requires MCMC). / EBM架构更灵活，但采样需要MCMC。
- Vs. GAN: EBM provides stable training and tractable (unnormalized) likelihood. GAN has fast sampling but unstable training and intractable likelihood. / EBM训练稳定，GAN采样快但训练不稳定且似然不可计算。
- Composability (组合性): $\text{S}_\theta = \text{vlimb}(\text{f}_\theta(x))$ can be combined with other models. / 多个EBM可组合使用，适用于零样本组合生成。

3. Key Formulas & Theorems / 重点公式与定理

3.1 Convolution / 卷积

- Formula: $\text{S}_\theta(x) = \text{vsum}_m \text{vsum}_n \text{H}_m \text{vlim}(\text{f}_\theta(x))$. / 卷积公式。
- Variables: H = kernel, F = input image, G = output image.
- Conditions: Kernel size determines neighborhood / 核大小决定邻域范围。

3.2 Gaussian Filter / 高斯滤波

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{1})\{\text{Z}(\theta)\}$. / 高斯滤波公式。
- Variables: S_θ = standard deviation (sigma), $\text{Z}(\theta)$ = normalization constant.
- Conditions: Larger S_θ means more blurring. / S_θ 越大越模糊。

3.3 Image Gradient / 图像梯度

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 图像梯度公式。
- Variables: S_θ = depth, F = focal length, B = baseline, d = disparity.
- Conditions: Rectified stereo pair / 校正后的立体对。

3.4 RANSAC Iterations / RANSAC迭代次数

- Formula: $\text{S}_\theta = \text{vfrac}(\text{log}(\text{1} - \text{p}))\{\text{log}(\text{1} - (1 - \text{p})^n)\}$. / RANSAC迭代次数公式。
- Variables: p = success probability, e = outlier ratio, S_θ = min sample size.
- Conditions: Assumes < 50% outliers / 假设外点比例 < 50%。

3.8 ELBO / 证据下界

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / ELBO公式。
- Variables: S_θ = variational distribution, $\text{Z}(\theta)$ = model params, V = variational parameters.
- Conditions: Equality when $\text{q}(z) = \text{p}(z)$. / 当变分分布等于后验时等号成立。

3.9 Change of Variables (Flow) / 变量替换 (标准流)

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 变量替换公式。
- Variables: f_θ is invertible transformation, S_θ is volume element.
- Conditions: f_θ must be invertible, S_θ continuous / f_θ 必须可逆， S_θ 连续且可微。

3.10 GAN Objective / GAN目标

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / GAN目标公式。
- Variables: S_θ = generator, D = discriminator, S_θ = prior on S_θ .
- Conditions: At optimum, equivalent to minimizing JSD / 最优时等价于最小化JSD数度。

3.11 Cross-Entropy Loss / 交叉熵损失

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 交叉熵损失公式。
- Variables: S_θ is score for class S_θ , S_θ is true label.
- Conditions: Most common classification loss / 最常用的分类损失。

3.12 Conv Output Size / 卷积输出尺寸

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 卷积输出尺寸公式。
- Variables: N = input size, F = filter size, P = padding.
- Conditions: Must be integer / 必须为整数。

3.13 Divergence / f-数度

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / Divergence公式。
- Variables: p = probability distributions, S_θ = convex generator function.
- Conditions: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 凸函数且分布相同。

3.14 Wasserstein Distance / Wasserstein距离

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / Wasserstein距离公式。
- Variables: W = Wasserstein distance, S_θ = 1-Lipschitz function.
- Conditions: Meaningful even for non-overlapping distributions / 即使分布不重叠也有意义。

3.15 EBM Probability / EBM概率

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / EBM概率公式。
- Variables: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 任意函数取指数后归一化。
- Conditions: $\text{Z}(\theta)$ generally intractable / 配分函数通常不可计算。

3.16 EBM Log-Likelihood Gradient / EBM对数似然梯度

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / EBM对数似然梯度公式。
- Variables: First term = positive phase, second term = negative phase.
- Conditions: Negative phase estimated via MCMC samples / 负相位通过MCMC样本估计。

3.17 Langevin MCMC / 朗之万MCMC

- Formula: $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / Langevin MCMC公式。
- Variables: S_θ = step size, $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$ = score function.
- Conditions: Converges to S_θ as $\text{S}_\theta \rightarrow 0$. / 当步长趋0，迭代次数趋无穷时收敛。

4. Typical Questions & Solutions / 典型问题与解答

Q1: Gaussian Filter Properties / 高斯滤波性质

- Problem (问题): If you convolve an image with a Gaussian of S_θ and then with a Gaussian of S_θ , what is the effective S_θ ?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 高斯滤波的性质。
- Problem (问题): Nyquist rate = S_θ times max frequency = 100 samples/pixel interval. Sampling rate must be S_θ times 100s.
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 奈奎斯特率。

Q2: Harris Corner / 奈奎斯特率

- Problem (问题): An image has maximum frequency 50 cycles/pixel. What is the minimum sampling rate to avoid aliasing?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 避免混叠的采样率。
- Problem (问题): For a matrix M , where the element matrix M has eigenvalues S_θ and S_θ , what is the effective S_θ ?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 矩阵的有效 S_θ 。

Q3: Harris Corner Classification / Harris角点分类

- Problem (问题): For a matrix M , where the element matrix M has eigenvalues S_θ and S_θ , what is the effective S_θ ?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / Harris角点分类。

Q4: Homography DOF / 单应性变换自由度

- Problem (问题): How many point correspondences are needed to solve for a homography?
- Solution (解答): A homography has 8 DOF. / 单应性变换有8个自由度。

Q5: RANSAC Rounds / RANSAC轮数计算

- Problem (问题): For computing a homography (s=4) with 30% outliers and p=0.99, how many RANSAC rounds?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / RANSAC轮数计算。

Q6: Stereo Depth Calculation / 立体深度计算

- Problem (问题): Given a stereo system with baseline $\text{B}=0.2\text{m}$, focal length $\text{f}=500$ pixels, and disparity $\text{d}=10$ pixels, what is the depth Z ?
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 立体深度计算。

Q7: Conv Layer Output Size / 卷积层输出尺寸

- Problem (问题): Input S_θ times S_θ times S_θ , 10 filters of size S_θ times S_θ times S_θ , stride 1, padding 0. What is the output size?
- Solution (解答): Output = $\text{S}_\theta - \text{f} + 1$. / 卷积层输出尺寸。

Q8: VAE ELBO Decomposition / VAE的ELBO分解

- Problem (问题): Write the ELBO in terms of reconstruction loss and KL divergence.
- Solution (解答): $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / VAE的ELBO分解。

Q9: Flow Model Likelihood / 标准流似然计算

- Problem (问题): In a normalizing flow with 3 invertible layers, write the log-likelihood.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 标准流似然计算。

Q10: GAN Optimal Discriminator / GAN最优判别器

- Problem (问题): Derive the optimal discriminator for fixed generator.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / GAN最优判别器。

Q11: Reparameterization Trick / 重参数化技巧

- Problem (问题): Why can't we directly backpropagate through sampling $z \sim \text{q}(z)$?
- Solution (解答): Sampling is a stochastic operation — gradients cannot flow through random nodes. Reparameterization: $\text{S}_\theta = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 重参数化技巧。

Q12: Change of Variables (1D) / 变量替换(一维)

- Problem (问题): If $\text{Z} = \text{U}[0, 2]$ and $\text{X} = \text{Z}^2$, find $\text{p}(x)$.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 变量替换(1D)。

Q13: NICE vs Real-NVP / NICE与Real-NVP区别

- Problem (问题): What is the key difference between NICE and Real-NVP coupling layers?
- Solution (解答): NICE uses a single coupling layer, while Real-NVP uses multiple. / NICE与Real-NVP区别。

Q14: MAF vs IAFF / 掩码回归与变自回归

- Problem (问题): When would you choose MAF over IAFF?
- Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse S_θ is parallel. Choose IAFF when you need efficient sampling (generation), since its forward S_θ is parallel. / MAF适合训练，IAFF适合生成。

Q15: Camera Projection Matrix / 相机投影矩阵

- Problem (问题): Write the full projection from world coordinates to pixel coordinates.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 相机投影矩阵。

Q16: WGAN vs. Original GAN / WGAN与原始GAN对比

- Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?
- Solution (解答): Wasserstein distance provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / WGAN的优势。

Q17: CycleGAN Cycle Consistency / CycleGAN循环一致性

- Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
- Solution (解答): Without cycle consistency, the adversarial losses alone cannot guarantee that S_θ is invertible. Cycle consistency enforces that S_θ is a valid transformation. / CycleGAN循环一致性的必要性。

Q18: MAF vs IAFF / 掩码回归与变自回归

- Problem (问题): When would you choose MAF over IAFF?
- Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse S_θ is parallel. Choose IAFF when you need efficient sampling (generation), since its forward S_θ is parallel. / MAF适合训练，IAFF适合生成。

Q19: Camera Projection Matrix / 相机投影矩阵

- Problem (问题): Write the full projection from world coordinates to pixel coordinates.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 相机投影矩阵。

Q20: WGAN vs. Original GAN / WGAN与原始GAN对比

- Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?
- Solution (解答): Wasserstein distance provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / WGAN的优势。

Q21: CycleGAN Cycle Consistency / CycleGAN循环一致性

- Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
- Solution (解答): Without cycle consistency, the adversarial losses alone cannot guarantee that S_θ is invertible. Cycle consistency enforces that S_θ is a valid transformation. / CycleGAN循环一致性的必要性。

Q22: MAF vs IAFF / 掩码回归与变自回归

- Problem (问题): When would you choose MAF over IAFF?
- Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse S_θ is parallel. Choose IAFF when you need efficient sampling (generation), since its forward S_θ is parallel. / MAF适合训练，IAFF适合生成。

Q23: Camera Projection Matrix / 相机投影矩阵

- Problem (问题): Write the full projection from world coordinates to pixel coordinates.
- Solution (解答): $\text{S}_\theta(x) = \text{vfrac}(\text{f}_\theta(x))\{\text{Z}(\theta)\}$. / 相机投影矩阵。

Q24: WGAN vs. Original GAN / WGAN与原始GAN对比

- Problem (问题): Why is Wasserstein distance preferable to Jensen-Shannon divergence as a GAN training objective?
- Solution (解答): Wasserstein distance provides a meaningful, continuous gradient even when distributions don't overlap. The loss correlates with sample quality, enabling more stable training. / WGAN的优势。

Q25: CycleGAN Cycle Consistency / CycleGAN循环一致性

- Problem (问题): Why is cycle consistency necessary in CycleGAN? What happens without it?
- Solution (解答): Without cycle consistency, the adversarial losses alone cannot guarantee that S_θ is invertible. Cycle consistency enforces that S_θ is a valid transformation. / CycleGAN循环一致性的必要性。

Q26: MAF vs IAFF / 掩码回归与变自回归

- Problem (问题): When would you choose MAF over IAFF?
- Solution (解答): Choose MAF when you need efficient density evaluation (training by MLE), since its inverse S_θ is parallel. Choose IAFF when you need efficient sampling (generation), since its forward S_θ is parallel. / MAF适合训练，IAFF适合生成。

Q18: EBM Gradient Interpretation / EBM梯度解释

- Problem (问题): Explain the positive phase and negative phase in EBM maximum likelihood training.

- Solution (解答):
 - Positive phase (正相位): Pushes up $\text{f}_\theta(x)$ at the training data location, making training data more probable. / 在训练数据位置推高 $\text{f}_\theta(x)$ ，提高训练数据概率。
 - Negative phase (负相位): Pushes